

LOCAL_AI ENGINE

A/D Context v20.10

1.

LOCAL_AI_ENGINE A/D Context v20.10 Pine Script v5 TradingView. (Accumulation/Distribution) , . , , , , .

: Chaikin AD Williams AD. Chaikin AD Money Flow Multiplier (MFM), close range (high - low). Williams AD close , . EMA-, CMF (Chaikin Money Flow), ATR .

v20.10 : Hidden () , Volume Filter Divergence Strength Score (0-10) . 3 Preset Auto TF , .

2.

2.1. Flow Core

. Money Flow Multiplier (MFM) Money Flow Volume (MFV). MFM close : ((close - low) - (high - close)) / range. MFV = MFM x volume. ta.cum() Raw Flow . EMA (A/D Smoothing EMA, 21), Smoothed .

CMF (Chaikin Money Flow) MFV Volume CMF Length (20). CMF : (0.15+), (-0.15-). ATR (Average True Range) 14 volatility .

2.2.

flowNorm = smoothed / fScale, fScale = EMA(|rawFlow|, regimeLB). : flow_z flow_pct. flow_z z-score: (smoothed - mean) / stdev . flow_z > 2 , flow_z < -2 . flow_pct (0% - 100%), smoothed high/low CMF Length . flow_pct 0% - , 100% - .

3.

overlay=false, .. price chart . 6 , background . .

Plot			
Raw Flow		1	MFV (Money Flow Volume). . trend.
Smoothed		2	EMA(rawFlow, 21). . divergence.
CMF		2	Chaikin Money Flow. 0 . CMF > 0 , CMF < 0 .
flow_z		1	Z-score smoothed. [-3, +3]. flow_z > 2 , flow_z < -2 .
flow_pct		1	smoothed . [0, 100%]. 0% = , 100% = .
Vol Ratio		1	Volume / SMA(volume, 20). Volume Filter ON. volMinRatio (0.50).

Settings Visual Show CMF / Show flow_z & flow_pct . Show Bias Background bias .

3.1. Zero-line Background

(hline 0) gray 80% . background bias: (green 90%) bias = bullish, (red 90%) bias = bearish, (gray 95%) neutral. , , .

4. (Info Table)

2 x 18 position.top_right . barstate.islast, .. 18 . .

Row		
profile	Preset 1/2/3/Manual (auto)	. Auto TF Preset (15m, 1H, 4H). - Manual.
TF / pivLR	15 8/8	Pivot Left / Pivot Right. : 8/8 15m, 10/10 4H.
div type	Regular / Hidden / Both	. Regular - , Hidden - , Both - .
bias	bullish / bearish / neutral	. : Flow Slope (1.0) + CMF Score (1.5) + Regime (1.0). >= 2.0 = bullish, <= -2.0 = bearish.
confirm	strong_bull / strong_bear / weak / none	bias . strong_ CMF > strong_ + slope > 0 + break high/low. weak - CMF > weak_. none - .
divergence	bullish / bearish / none	- . last bar . "x" .
regime	bull / bear / flat / transition	flow . bull = N , bear = N . flat = . transition = .
quality	high / medium / low	0-7. high >= 6: bias + strong confirm + divergence + regime + strength >= 5. medium >= 3. low < 3.
div_str	X.X/10	Divergence Strength Score . Price (max 3) + Flow (max 3) + ATR (max 2) + Volume (max 2). n/a = .
cmf	#####	Chaikin Money Flow . CMF > 0 , CMF < 0 . .
slope	format.mintick	Smoothed : smoothed - smoothed[1]. slope > 0 flow , slope < 0 flow .
raw_flow	format.mintick	Raw Flow (cumulative MFV). . .
smoothed	format.mintick	EMA(rawFlow, 21). . .
flow_z	###	Z-score smoothed. [-3, +3]. 0.
vol_ratio	### / off	Volume / SMA(volume, 20). Volume Filter ON. red vol_ratio < volMinRatio.
flow_pct	###%	smoothed . 0% = , 100% = .
cooldown	N	. . 12-20 .
comment		. .

: color-code: bullish = green, bearish = red, neutral = gray. .

5.

5.1. Regular Hidden

: Regular () Hidden (). Regular . Hidden .

Regular Bull	Price: < ; Flow: >	: price , flow . .
Regular Bear	Price: > ; Flow: <	: price , flow . .
Hidden Bull	Price: > ; Flow: <	: price , flow . . .
Hidden Bear	Price: < ; Flow: >	: price , flow . . .

Detection Mode (Regular / Hidden / Both). Hidden labels Distinguish Hidden in labels = true, label HID (BUY DIV HID). : .

5.2. Divergence Strength Score (0-10)

v20.10 Strength Score 0 10. 4 :

Price (max 3)	priceDelta / priceRef / 0.05	3. 5% 3 .
Flow (max 3)	flowDelta / flowRef / 0.10	3. 10% 3 .
ATR (max 2)	priceDelta / ATR / 0.5	2. 0.5 ATR 2 .
Volume (max 2)	volRatio / 1.0	2. volRatio >= 1.0 (volume > SMA) 2 .

Min Strength to Show (divStrMin) . divStrMin = 0.0 . 3.5 - 4.0 . div_str info table strength >= 5.0 quality +1 .

5.3.

follow-through () Req Confirm. Req Confirm = true, divergence "pending" follow-through confirm bars (6). follow-through: close > high[1] close > open body/range >= 0.2 (for bull), close < low[1] close < open body/range >= 0.2 (for bear). follow-through , .

Cooldown Bars , . 12-20 . cooldown , .

5.4.

live-divergency label . label style :

Label		
BUY DIV		Regular Bull Divergence. .

Label		
BUY DIV HID [X.X]		Hidden Bull Divergence. HID hidden, [X.X] = strength score.
SELL DIV		Regular Bear Divergence.
SELL DIV HID [X.X]		Hidden Bear Divergence.
BUY DIV?		pending bull div - follow-through.
SELL DIV?		pending bear div - follow-through.
bull? / bear?		pending follow-through . .
b? / s?	dim	historical unconfirmed (dim) . .
B / S	dim	historical confirmed (Text style).
x		"x" . "x" bar_index-1 .

5.5. Divergence Lines

Show Active Div Zone Line = true live- . (green, solid, width 2) (red, solid, width 2). Max Line Length (bars) (80) gray (invalidated). historical- dashed width 1. "x" Show History Div End marker .

6. Regime Bias

6.1. Regime Detection

Regime flow . : slopePct (slope lookback regimeLB), rangePct (high/low), consecutive bars (bullCnt / bearCnt) Regime Confirm Bars. :

Regime		
bull	bullCnt >= N, slopePct > 0.15, rangePct > 0.20, smoothed > lagFlow	N . flow .
bear	bearCnt >= N, slopePct < -0.15, rangePct > 0.20, smoothed < lagFlow	N . flow .
flat	slopePct < 0.15	. CMF slope .
transition		. regime .

flat- : slope < 0 CMF < weak_ regime bear, slope > 0 CMF > weak_ regime bull. flat .

6.2. Bias Scoring

Bias weighted score 3 : Flow Slope, CMF Score Regime. biasThresh (2.0) .

	Score	
Flow Slope (wFlow = 1.0)	slope > 0 = +1.0, slope < 0 = -1.0	EMA smoothed .
CMF Score (wCmf = 1.5)	CMF > strong = +1.5, CMF > weak = +0.5, CMF < -strong = -1.5, CMF < -weak = -0.5	CMF . strong/weak TF .
Regime (wRegime = 1.0)	bullReg = +1.0, bearReg = -1.0	regime (6.1).

bScore = fScore x wFlow + cScore x wCmf + rScore x wRegime. bScore >= biasThresh (2.0) = bullish, bScore <= -biasThresh = bearish, = neutral. CMF TF: 1m CMF strong = 0.02, 1H CMF strong = 0.05, 4H CMF strong = 0.06.

6.3. Confirmation System

bias confirmation . : strong weak.

Confirm		
strong_bull	bias=bullish + CMF > strong_ + slope > 0 + close >= highest(high, brLen)	break recent high. .
strong_bear	bias=bearish + CMF < -strong_ + slope < 0 + close <= lowest(low, brLen)	break recent low. .
weak	bias + CMF > weak_ + slope > 0 (bull) CMF < -weak_ + slope < 0 (bear)	bias CMF , break.
none		.

7. Volume Filter

v20.10 Volume Filter . Enable Volume Filter = true . : volume / SMA(volume, volSmaLen). volRatio >= volMinRatio (0.50), . volRatio < volMinRatio, divergence .

Volume Ratio Vol Ratio (aqua, stepline) Vol Min (aqua, dotted). vol_ratio info table red volRatio < volMinRatio. .

8. Quality Comment

8.1. Quality Score

Quality 0-7, :

Bias present (bullish/bearish)	+1
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Strong confirmation	+2
Weak confirmation	+1
Divergence present	+1
Regime = bull/bear	+1
Strength >= 5.0	+1

: qs >= 6 = high, qs >= 3 = medium, qs < 3 = low. quality info table color-code: high = green, medium = orange, low = gray. high quality .

8.2. Comment

Comment info table . :

Comment	
div candidate - await follow-through	pending divergence .
countertrend flow vs bias	flow bias (: bias=bearish slope>0 bias=bullish slope<0).
bullish divergence	bull divergence .
bearish divergence	bear divergence .
bullish vol confirm	strong_bull confirmation.
bearish vol confirm	strong_bear confirmation.
bullish vol pressure	bias=bullish confirmation.
bearish vol pressure	bias=bearish confirmation.
neutral	.

9.

5 alertcondition. TradingView .

Alert		
BUY DIV	Alert: BUY DIV = true	bull divergence .
SELL DIV	Alert: SELL DIV = true	bear divergence .
Bull Div Candidate	Alert: DIV Candidate = true	pending bull div follow-through.
Bear Div Candidate	Alert: DIV Candidate = true	pending bear div follow-through.
DIV (any)	BUY DIV SELL DIV	divergence.

10.

10.1. XAUUSD (Gold)

Preset	TF	Piv L/R	Min Price %	Min Flow %	Min ATR	Cooldown n	Confirm
Preset 1 (15m)	15	8/8	0.08	0.08	0.35	12	false
Preset 2 (1H)	60	8/8	0.15	0.10	0.40	15	false
Preset 3 (4H)	240	10/10	0.25	0.12	0.50	20	true

10.2. BTCUSD (Bitcoin)

Preset	TF	Piv L/R	Min Price %	Min Flow %	Min ATR	Cooldown n	Confirm
Preset 1 (15m)	15	10/10	1.0	0.12	0.50	12	false
Preset 2 (1H)	60	8/8	1.5	0.10	0.45	15	false
Preset 3 (4H)	240	10/10	2.5	0.14	0.55	20	true

: BTC XAUUSD ~10x, Min Price % . Flow % . 4H Preset Req Confirm = true follow-through .

11.

11.1.

, quality, confirmation regime. .

BUY DIV [7.0+] + strong_bull + regime = bull		. break recent high. .
BUY DIV [5.0-7.0] + weak + quality = high		. CMF break. .
BUY DIV HID + quality >= medium		hidden . . .
SELL DIV [7.0+] + strong_bear + regime = bear		. break recent low. .
SELL DIV [5.0-7.0] + weak + quality = high		. CMF break. .
SELL DIV HID + quality >= medium		hidden . . .
div candidate + quality = low		. . follow-through .

11.2.

:...:

1. info table: bias, confirm, regime, quality, div_str comment. .
2. Strong Score ($\geq 7/10$) . score 3-5 .
3. quality = high + confirmation = strong .
4. quality = low comment = neutral . .
5. pending (BUY DIV?) : follow-through . .
6. Hidden (HID) : , trend-continuation, trend-reversal.
7. Volume Filter ON vol_ratio < volMinRatio . .
8. "x" . .
9. Countertrend flow vs bias (comment): . .
10. Cooldown: . , cooldown .

11.3.

: MTF (Multi-Timeframe) . 3 Preset . : 4H Preset , 1H , 15m . 3 TF , . MTF- . 4H divergence 1H/15m .